LESSON PLAN

WEAPON GUIDANCE (19)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	09
Total No. of Lesson Plans	22	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Explanation on Wire routing and protection hazards.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Explanation on Wire routing and protection hazards.
5. Specific Objectives	19.1.1.1Trainee will be able to explain aircraft wiring and Causes of wiring degradation.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of IB diagrams regarding wire routing paths; Instructor supervision on identifying protection hazards in schematic layouts.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of aircraft wiring?
(b) List three common causes of wiring degradation.
(c) Why is proper wire routing critical for aircraft safety?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Fundamentals of Aircraft Wiring Systems • Standard Wire Routing Principles • Types of Wire Protection Methods • Common Routing Hazards Identification • Impact of Improper Routing on Systems	L & Ds	PPT, IB	15 min
(b)	• Detailed Analysis of Routing Hazards • Protection Mechanisms Against Degradation • Case Studies of Wiring Failures • Preventive Maintenance Procedures • Inspection Techniques for Wire Integrity	L & Ds	PPT, PDF	20 min
(c)	• Summary of Wire Routing Best Practices • Review of Protection Hazard Mitigation • Linking Routing to System Reliability • Key Takeaways for Maintenance Crew • Transition to Wire Chafing Concepts	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how improper wire routing leads to system degradation.
(b) Describe two specific protection hazards and their mitigation strategies.

12. Home Assignment and Follow Up	Prepare a brief note on how the wire routing principles discussed in this lesson apply to the wire chafing causes covered in LP 10.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Wire chaffing and various causes of wire chafing.	Refer to the master syllabus reference list	LP 10

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min